Univariate :

1) Central Dedency

```
[11]: for colName in quan:
                descriptive[colName]["Mean"]= dataset[colName].mean()
                descriptive[colName]["Median"]=  dataset[colName].median()
                descriptive[colName]["Mode"]=  dataset[colName].mode()[0]
      descriptive
```

[11]:

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Mean	108.0	67.303395	66.333163	66.370186	72.100558	62.278186	288655.405405
Median	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Mode	1	62.0	63.0	65.0	60.0	56.7	300000.0

In this image, Means almost close for Sl_no, ssc_p, degress_p, etest_p, mba_p. but salary only having high level difference in mean and median. So, salary having outlier.

2) Percentile

```
for colName in quan:
            descriptive[colName]["Mean"]= dataset[colName].mean()
            descriptive[colName]["Median"]=  dataset[colName].median()
            descriptive[colName]["Mode"]=  dataset[colName].mode()[0]
            descriptive[colName]["Q1:25%"]=  dataset.describe()[colName]["25%"]
            descriptive[colName]["Q2:50%"]= dataset.describe()[colName]["50%"]
            descriptive[colName]["Q3:75%"]= dataset.describe()[colName]["75%"]
            descriptive[colName]["99%"]= np.percentile(dataset[colName],99)
            descriptive[colName]["Max"]=  dataset.describe()[colName]["max"]
descriptive
```

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Mean	108.0	67.303395	66.333163	66.370186	72.100558	62.278186	288655.405405
Median	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Mode	1	62.0	63.0	65.0	60.0	56.7	300000.0
Q1:25%	54.5	60.6	60.9	61.0	60.0	57.945	240000.0
Q2:50%	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Q3:75%	161.5	75.7	73.0	72.0	83.5	66.255	300000.0
99%	212.86	87.0	91.86	83.86	97.0	76.1142	NaN
Max	215.0	89.4	97.7	91.0	98.0	77.89	940000.0

```
import numpy as np
```

In this image, Q1, Q2, Q3, 99%, max are different type of location for splitting data percentage.

For Examples, Consider as ssc_p

Q1 = 25% , Q2= 50% = 60.6 -67.0 = overall 7% percentage difference from Q1 to Q2

Q2 = 50% , Q3= 75% = 67.0 -75.7 = overall 8% percentage difference from Q2 to Q3

Q3 = 50% , 99% = 75.7-87.0 = overall 12% percentage difference from Q3 to 99%

99% , Q4= 100% or MAX = 87.0 -89.4 = overall 2% percentage difference from 99% to 100%

Based on above data each percentage wise categorize the data percentage.